

WELLS, ME, USA

WMO: 997707

Lat: 43.320N

Lon: 70.563W

Elev: 6

StdP: 101.25

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -15.7	(c) -13.1	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 11.6	(k) -2.8	(l) 10.5	(m) -2.8	(n) 3.6	(o) 310	(p) 0.541

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 7.3	(c) 29.3	(d) N/A	(e) 26.9	(f) N/A	(g) 25.2	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 3.7	(p) 280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 10.0	(b) 8.7	(c) 7.7	(d) DB	(e) -18.1	(f) 33.1	(g) 2.2	(h) 2.2	(i) -19.7	(j) 34.7	(k) -21.0	(l) 36.0	(m) -22.2	(n) 37.2	(o) -23.8	(p) 38.8	(q) N/A
(4)			(r) WB	(s) N/A	(t) N/A	(u) N/A	(v) N/A	(w) N/A	(x) N/A	(y) N/A	(z) N/A	(aa) N/A	(ab) N/A	(ac) N/A	(ad) N/A	(ae) N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.7	-3.5	-2.2	1.3	6.9	12.4	16.7	20.4	19.5	16.0	10.7	5.0	0.6
	DBStd	9.06	5.67	4.69	4.50	3.59	3.83	3.44	2.89	2.46	3.55	3.62	4.31	4.61
	HDD10.0	1641	419	341	271	106	16	1	0	0	1	34	159	293
	HDD18.3	3686	677	574	527	343	188	73	10	15	89	238	401	551
	CDD10.0	1168	0	0	3	13	91	200	322	295	179	56	9	0
	CDD18.3	172	0	0	0	0	6	22	73	52	18	1	0	0
	CDH23.3	868	0	0	1	4	73	133	377	191	80	10	0	0
	CDH26.7	228	0	0	0	0	22	36	109	39	20	1	0	0
(14) Wind	WSAvg	3.4	3.6	3.7	4.1	3.9	3.5	3.2	2.9	2.8	2.9	3.4	3.4	3.3
(15) Precipitation	PrecAvg	1190	86	82	110	102	94	103	96	88	100	121	101	107
	PrecMax	1637	168	215	388	250	372	319	193	287	268	375	199	171
	PrecMin	880	14	30	22	7	18	28	30	7	36	9	34	35
	PrecStd	216	30	38	71	57	69	61	47	54	57	83	38	32
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.7	11.7	19.7	22.7	29.9	30.7	32.9	30.6	29.3	24.7	17.9	13.6
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	8.9	7.9	13.0	17.8	25.0	26.8	29.6	27.4	25.5	19.5	15.2	10.7
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	7.2	5.7	9.2	14.7	21.2	24.3	27.1	25.4	22.9	17.2	13.1	9.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	5.1	4.2	7.1	12.5	17.9	21.9	25.2	23.7	21.3	15.8	11.7	7.5
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(35) Mean Daily Temperature Range	5% DB	MDBR	8.2	8.1	7.5	7.7	7.7	7.0	7.3	7.3	7.8	7.7	7.6	6.8
		MCDBR	9.6	9.6	10.2	11.7	13.8	11.3	10.6	9.8	9.5	9.2	9.4	8.4
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(40) Clear-Sky Solar Irradiance	taub		0.291	0.301	0.325	0.357	0.394	0.445	0.461	0.429	0.378	0.336	0.316	0.296
	taud		2.473	2.458	2.464	2.418	2.357	2.245	2.211	2.289	2.409	2.523	2.541	2.535
	Ebn at Noon		851	902	919	909	879	832	813	828	849	850	815	814
	Edh at Noon		72	86	97	110	121	136	139	125	102	81	68	63
(44) All-Sky Solar Radiation	RadAvg		1.70	2.57	3.66	4.62	5.16	5.54	5.84	5.30	4.26	2.76	1.83	1.37
	RadStd		0.11	0.20	0.23	0.49	0.58	0.46	0.39	0.26	0.24	0.25	0.16	0.15

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (62 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+81	+30

Nomenclature: See separate page